Using the details from the problem statement (section 11.3), design the DCC contract diagram.

11.5 Developing the smart contract

Use the contract diagram, develop the smart contract, and do a quick test in the Remix IDE. After a successful test in Remix, open the DCC-Dapp codebase to review the smart contract in the DCC-contract directory. I explain these steps in detail next.

11.5.1 Data structures

Listing 11.1 shows the data structure that stores the data about a student's courses, the modifiers, the events, and the functions. The names of the items are self-explanatory, and they follow camel-case notation.

11.5.2 Events

There are six event definitions, one for each category of courses. An event is emitted when the courses in a category are completed. The code also includes two more events: one emitted when the GPA is satisfied and one that shows the current GPA value.

11.5.3 Modifiers

There are two modifiers. `checkStudent` allows a student to register only once, and `validStudent` enforces that only a valid student is permitted to add courses. In other words, before